

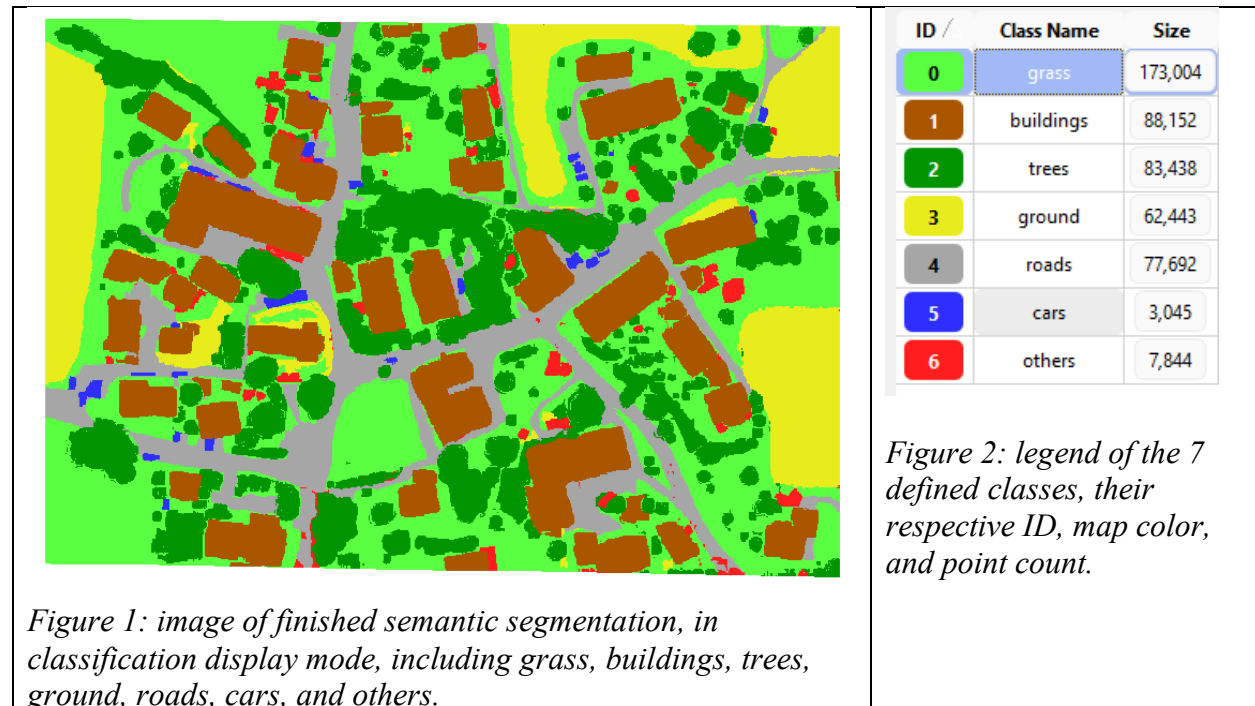
LSGI3262 Laser Scanning Lab 5

Student name: [REDACTED]

Student ID: [REDACTED]

Task 1 – Final Semantic Segmentation and Labeling in Classification Mode

This lab assignment focused on a semantic segmentation of a laser scanning point cloud dataset representing a typical suburban neighborhood. The dataset includes diverse urban elements, and the goal was to complete semantic labeling of the `sem_1b1.las` file using 3DMax, defining at least 4 custom categories. A combination of manual and semi-supervised segmentation was carried out and 7 semantic classes were defined.



Defined classes and description

Class ID	Class Name	Color	Description
0	Grass	Lime	Low, homogeneously light-colored, vegetation, lawns, garden areas
1	Buildings	Brown	Houses, roofs
2	Trees	Green	High, dark, clustered vegetation, tree canopies.
3	Ground	Yellow	Bare earth grounds, unpaved non-asphalt or concrete surfaces, with little vegetation.
4	Roads	Gray	Driveways, parking areas

5	Cars	Blue	Vehicles (parking or driving)
6	Others	Red	Noise points, fences, poles, unidentified or miscellaneous objects

Task 2 – Detailed Process Description

Step 1: Semi supervised segmentation for homogeneous, low noise values (one click segmentation/Segment Anything Mode)

For large, homogeneous regions with relatively low internal variation and noise, the Segment Anything Mode (one-click segmentation tool) in 3DMas was used. This tool propagates the selection from a single clicked point across similar neighboring points based on geometric and radiometric similarity (e.g., planar surfaces, consistent height, density).



Figure 1: Segment Anything Mode in 3DMas



Figure 2: selecting a building after clicking on the center of the roof

This tool is particularly effective for entire building roofs, large ground, tree, or grass patches, and continuous road segments. In practice, one click often captured over 95% of the points of the target feature, greatly reducing the data processing time compared to fully manual methods.

This semi-supervised method tends to work in the highest accuracy in homogeneous intensity regions. However, minor leakage might occur near object boundaries (e.g. roof edges touching tree edge), and requires subsequent manual correct to include/exclude the points.

Step 2: Manual segmentation for heterogeneous or high noise values

Manual polygon selection was carried out for areas that have heterogeneous intensities, shadowed areas, occlusions, high noise areas, or varying point densities (too sparse for Segment Anything Mode to register).

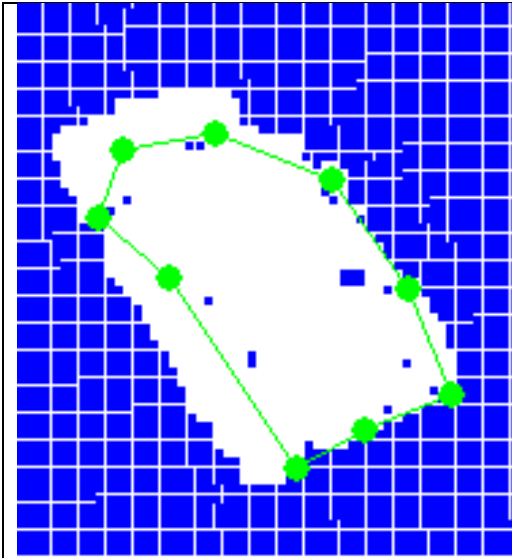


Figure 3: There might be some noise points after applying Segment Anything Mode, therefore it is required to use Polygon Selection Tool to select the leftover unregistered noises.

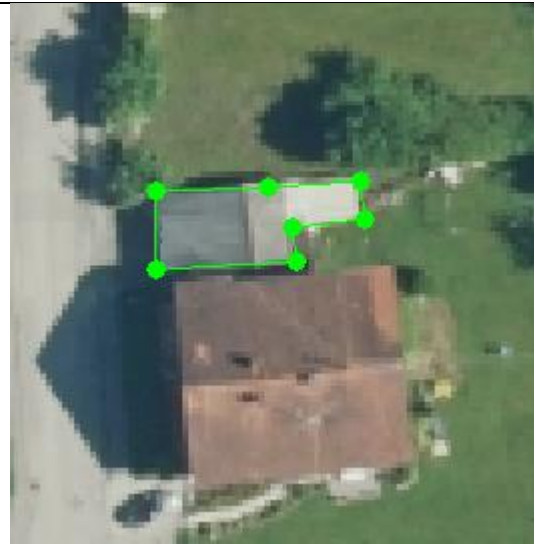


Figure 4: Using Polygon Selection Tool to select the entirety (sometimes partial) of individual features

There are some classes that frequently require manual segmentation. For example, noise points inside a building, building edges close to trees or grass, sparse trees might blend into the grass, and small miscellaneous objects, such as fences, poles, and unidentifiable objects.

The polygon selection tool requires precise manual control over the class boundary, which usually gives the highest accuracy, ensuring only the intended points were captured without including adjacent objects. Selected points were then assigned to the correct class. Although Step 1 (one click segmentation) was used in the initial attempt and might reduce labor, manual selection remains necessary and important to achieve clean boundaries and correct previous misclassifications from the Segment Anything Tool.

Step 3: Merge point clusters into the same class

After initial segmentation, individual point clusters were often set to a different class. This lets the user double confirm the selected point clouds are intended and can be merged into the same class. The unintended points can be removed by only viewing in classification mode. During merging, a temporary uniform color were assigned to each class and can be changed later. Merged ensured that all points of the same type will share the same final class ID and display color in classification display mode.

After merging to the corresponding class ID, the points can be further smoothed and cleaned in case of any noises or unintended points included. Post-merge smoothing removed small outlier

clusters or jagged edges, enhancing visual smoothness in classification mode without altering semantics. This verification step was crucial to ensure the final dataset was coherent and ready for isolated class review.

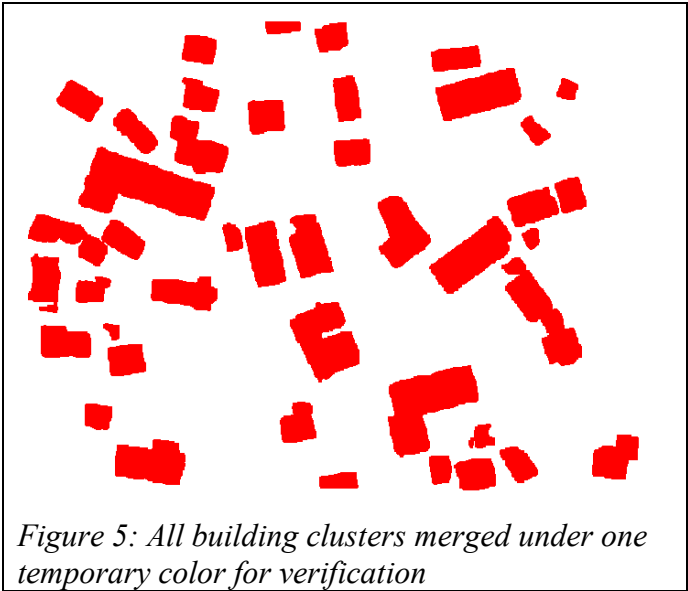


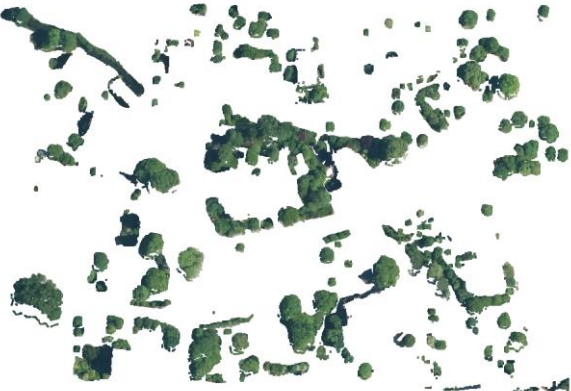
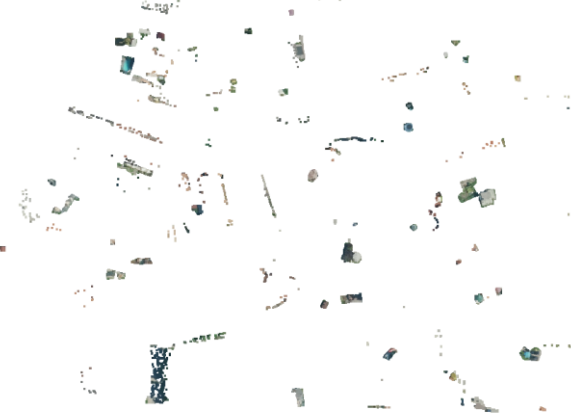
Figure 5: All building clusters merged under one temporary color for verification

Step 4: Visually inspect results in RGB mode

Below shows the point clusters in RGB mode in respect to their classes.

Reviewing each class in isolation (showing only one class at a time) confirmed high segmentation quality: grass patches showed uniform coverage without road/building intrusion; buildings appeared as complete, non-fragmented volumes; trees displayed distinct canopies without ground mixing; roads maintained continuity; cars were compact and separate; others contained only intended small features/noise. Minimal leakage or holes across views indicated successful boundary control and merging.

Grass	Buildings
Trees	Ground

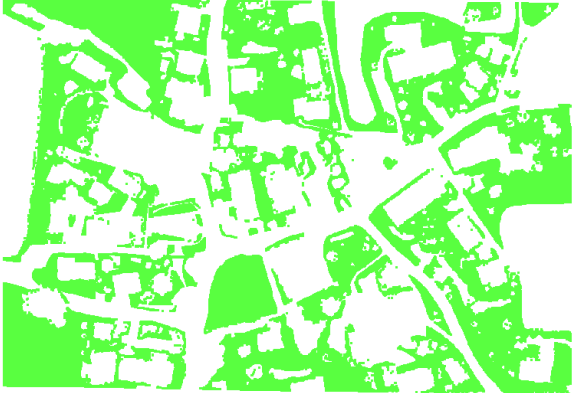
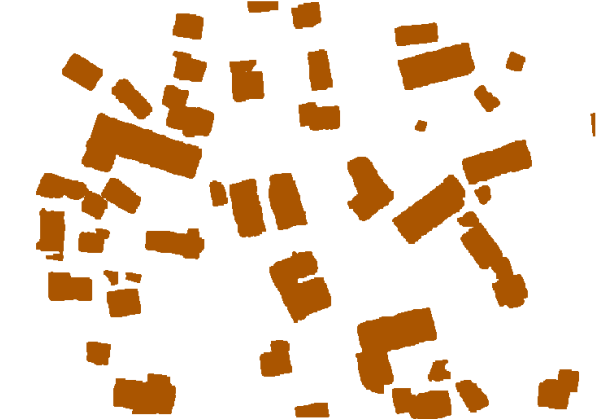
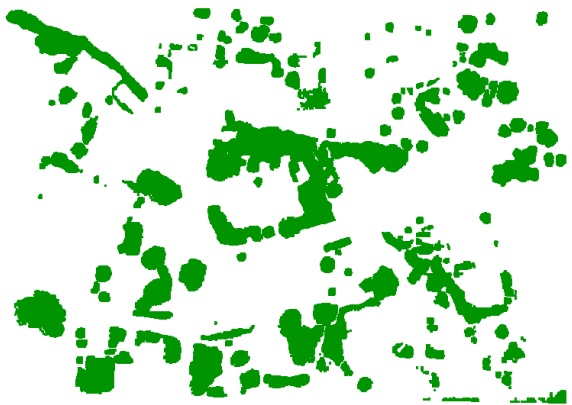
	
Roads	Cars
	
Others	
	

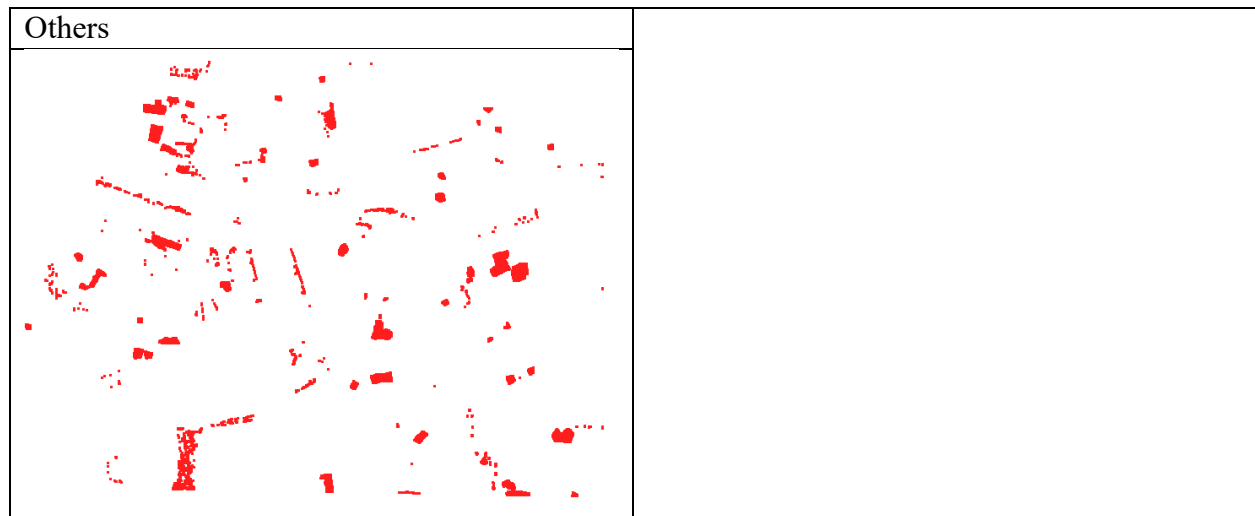
Step 5: Segmented Results in Classification Display Mode

The fully labeled point cloud is displayed in Classification mode, with each class shown in its assigned color. The result clearly separates semantic elements: lime grass areas, brown building

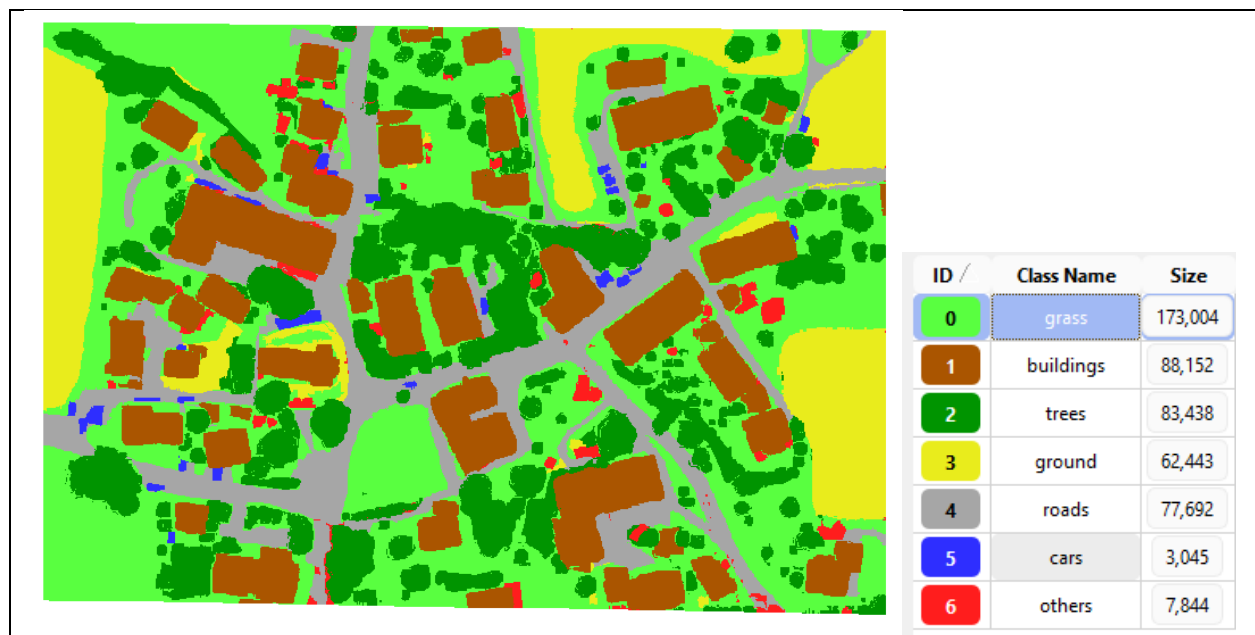
volumes, green tree canopies, yellow bare ground, gray roads, blue vehicles, and red miscellaneous objects.

Below are the isolated class views, in classification mode and showing one class at a time to further confirm the quality of the segmentation.

Grass	Buildings
	
Trees	Ground
	
Roads	Cars
	



Step 6: Final Result in Classification Display Mode



Conclusion

This lab assignment used a combination of two semantic segmentation methods, manual and semi-supervised segmentation. The semi-supervised workflow successfully completed the initial semantic labeling, efficiently segmenting the homogeneous regions using one click. On the other hand, manual polygon selection ensured precision and accuracy in complex and noisy areas, cleaning the residual noise from Segment Anything Mode. Together, they segment the point clouds into clusters of individual ground features. Merging and further cleaning under the same classification results in clean and interpretable point cloud in classification display mode.

This exercise highlights the strengths and limitations of manual and semi-supervised segmentation in 3DMas. It can be made with high accuracy and detailed segmentation, but the tradeoff is that it could be time-consuming for large or highly detailed scenes. The detailed 7 classes exceeds the minimum requirement, and provides richer semantic information suitable for urban analysis tasks. The process of segmentation also aligned with the lecture content, and this exercise demonstrates that segmentation can have practical applications for real-world suburban laser scanning data.